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ORIGINAL RESEARCH

# XGBoosting Cricket: Enhancing Predictive Modeling for Twenty20 Match Results Using Machine Learning and Statistical Techniques

Akhil Tyagi[1] · Amandeep Kaur[1] · Aryan Kamboj[1] · Chayandeep Chaulia[1] · Gandharv Mohan[1] · Manpreet Singh[2]

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## Abstract

The emerging field of computer science research that combines Data Mining and Machine Learning in sports analytics, has revolutionized performance analysis and predictive modeling. This study presents a predictive model leveraging the XGBoost algorithm to forecast scores in Twenty20 (T20) cricket matches, with a comparative analysis against other popular regression algorithms such as Linear Regression, Decision Trees, and Random Forests. The model incorporates a comprehensive dataset of historical T20 match statistics, including player performance, match conditions, and venue changes. Rigorous preprocessing and feature selection techniques were employed to optimize model performance. The results demonstrate that the XGBoost algorithm outperforms other methods, achieving a Mean Absolute Error (MAE) of 1.73 on the test dataset. This margin of error highlights the model's precision in predicting T20 match outcomes, providing valuable insights for strategic decision-making in cricket analytics. The findings suggest that advanced machine learning techniques like XGBoost can significantly enhance predictive accuracy in sports analytics, paving the way for future research and practical applications in the field.

**Keywords** Machine learning · Random forest · XGBoost · Linear regression · Decision tree · Comparative analysis

## Introduction

The sport of cricket, tracing its roots back to the sixteenth century in England, has evolved into a multifaceted discipline with various formats, diverse playing standards, and fluctuating durations. Among the contemporary formats recognized by the International Cricket Council (ICC), Twenty20 (T20) stands out as a dynamic and captivating version of the game. In T20 cricket, each team has a single inning, constrained to a maximum of 20 overs, resulting in a fast-paced and thrilling spectacle. This format has garnered immense success owing to its short duration and the exhilarating moments it generates, leading to the organization of numerous annual tournaments at both domestic and international levels.

The popularity of T20 cricket has spurred a wealth of analyses focusing on individual and team performance, as well as predictions for future games, spanning across all formats of the game. Currently, strategic decisions in cricket are often informed by a combination of player experience, team composition, and what is commonly referred to as "cricketing sense." Amidst this backdrop, this research directs its attention to the T20 format, chosen for its widespread appeal and influence.

This study endeavors to employ Machine Learning principles to develop a sophisticated prediction system specifically tailored for the T20 format. Machine Learning, a transformative field within computer science, offers two prominent approaches—supervised and unsupervised machine learning. In the realm of supervised machine learning, the machine is trained using extensive datasets and the corresponding outcomes. This project harnesses various classification algorithms, including One Hot Encoder, Column Transformer, and Standard Scaler, alongside a comparative Analysis of Various Regression Algorithms. Additionally, this project employs the XGBoost Algorithm to make precise predictions regarding final scores, enhancing the accuracy and comprehensiveness of this predictive model.

✉ Amandeep Kaur
amandeep.gtbit@gmail.com

[1] Department of Information Technology, Guru Tegh Bahadur Institute of Technology, New Delhi, India

[2] Boston University, Boston, USA

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## Literature

Cricket, as one of the most popular sports globally, has witnessed a surge in interest and analysis with the advent of Twenty20 (T20) cricket. The dynamic and fast-paced nature of T20 matches has led researchers and data enthusiasts to explore the application of machine learning techniques for predicting match outcomes and, more specifically, predicting the final scores. The motivation for undertaking the project on T20 cricket score prediction stems from the growing importance of data-driven insights in the realm of sports analytics. The impetus for this research arises from the realization that predicting cricket scores, especially in the T20 format, is a challenging yet rewarding task. Accurate score predictions can provide valuable insights for teams, coaches, and fans alike. The fast-paced nature of T20 cricket, with its emphasis on aggressive batting and strategic bowling, adds an extra layer of complexity to the prediction task. The goal is to leverage machine learning algorithms to develop a robust and accurate model capable of predicting T20 cricket scores.

Sports analytics in cricket has witnessed a surge of interest in recent years, with researchers exploring diverse aspects of the game's dynamics, player performance, and predictive modeling. This review examines a few pivotal papers that have contributed significantly to the evolving field of cricket analytics. A summary of the few referenced papers is given in the following Table 1.

## Methodology

The ML-based approach employed that involves the utilization of machine learning principles. The fundamental requirements of an ML algorithm include the acquisition of a dataset, the training of said dataset utilizing the algorithm, and the subsequent testing of the model.

### Data Set Used

The dataset has been obtained from Kaggle [14]. With regard to score prediction, current research encompasses the compilation of data pertaining to all T20 matches played since 2000. The dataset comprises 76,015 rows and encompasses 15 columns. Application of feature selection techniques resulted in the identification of 8 pertinent features, of which 7 serve as input features and 1 functions as a target variable as given in Table 2. The chosen attributes encompass the batting team, bowling team, city, current score, balls left, wickets left, current run rate, runs in the previous 5 overs, and runs for score prediction. The initial step involves importing the dataset utilizing the pandas library. Using the OS module, all.yaml files have been transformed into readable files, facilitating the incorporation into the data frame for subsequent processing.

Following this, the dataset was further preprocessed by identifying null values, which were then replaced with the mean or median values of the corresponding column. Categorical data in the columns is converted into numerical values. Feature selection techniques are then employed to opt for the most suitable set of features.

### Feature Selection

The process of Feature Selection involves the selection of an optimal set of features from the input features set through the application of feature selection techniques. By eliminating redundant features, the dimension of the data is reduced, leading to potential improvements in both time and space complexity. The performance of the model is enhanced through feature selection, contributing to time and space savings.

Following this, the dataset underwent further preprocessing, involving the identification of null values, which are then replaced with the mean or median values corresponding to the respective column. Categorical data in the columns is converted into numerical values. Feature selection techniques are then employed to opt for the most suitable set of features.

In the next Step, the prediction model has been formulated, and the train-split-test method is been applied to segregate the data into training and testing datasets. Afterward, the prediction model has been undergone training with the training data.

### Algorithm

The potentials of existing data mining techniques considered relevant to the research were initially understood. The background information pertinent to the study was documented in this section.

With a focus on predictions, the research necessitates the application of supervised learning, involving the training of a model using both input and desired output labels. Two primary approaches within supervised learning are classification and regression.

In this T20 cricket score prediction project, various machine learning algorithms are employed to forecast T20 match scores based on historical data. Linear Regression serves as a foundational model, capturing linear relationships between features and target scores. Decision Trees are introduced to account for complex interactions among diverse cricketing factors, providing insights into feature importance. Random Forest, an ensemble method, mitigates

**Table 1** Summary of a few referenced papers

| S.No | Paper Name | Algorithm | Merits | Demerits |
|---|---|---|---|---|
| 1 | Score and Winning Prediction in Cricket through Data Mining [1] | 1.Linear Regression algorithm<br>2.Naive Bayes classifier | 1.Accuracy of LR is 20% greater than that of the CRR method<br>2.Outcome accuracy also increases with an increase in the overs | 1.The prediction is done for the ODI matches<br>2.Other factors like the toss, ranking of the team, venue, home team advantage are not considered |
| 2 | Cricket MatchOutcomePrediction UsingTweets and Prediction of the Man of the Matchusing Social Network Analysis: Case Study Using IPL Data [2] | 1.Naive Bayes classifier<br>2.Support Vector Machine(SVM)<br>3.K-Nearest Neighbor method<br>4.Random Forest method<br>5.Logistic Regression | 1.Sentiment analysis has an accuracy of up to 85%<br>2. Tweet-based and Mixed model has 89% accuracy<br>3.Natural parameter-based model has accuracy up to 83% | 1.Sometimes prediction using tweets may go wrong as the game can change very quickly within the span of 2 overs as the span for the tweet-based model is 10 overs |
| 3 | Player'sPerformancePrediction in ODI Cricket UsingMachineLearning Algorithms [3] | 1.Linear Regression<br>2.SVM with linear kernel<br>3.SVM with polynomial kernel | 1.SVM with linear kernel has greater accuracy as compared to LR and SVM with the polynomial kernel | 1.Considers only ODI matches<br>2.Accuracy is less |
| 4 | OutcomePrediction ofODICricket Matches using Decision Trees and MLP Networks [4] | 1.Multilayer Perceptron Network<br>2.Decision Trees | 1.MLP Networks are capable to learn & train in real-time using partial fitting property<br>2.Decision trees can handle both numeric as well as categorical data and also multi-output problems | 1.MLP Networks are highly sensitive for feature scaling<br>2.Decision Trees are usually very unstable and even small modifications in the data might lead to an entirely different tree being generated |
| 5 | Money Ball—DataMining on Cricket Dataset [5] | 1.Naive Bayes classifier<br>2.SupportVector Machine(SVM)<br>3.K-Nearest Neighbor method<br>4.Random Forest method | 1.Random Forest algorithm has the most accurate result | 1.Every team has different features/attributes and using those attributes the best scenarios are considered |
| 6 | Cricket Squad Analysis using multiple RandomForest Regression [6] | 1.Linear Regression<br>2.SVR (Support Vector Regression)<br>3.Decision Tree<br>4.Random Forest | 1.Provides a mathematical approach to select the players<br>2.RMSE value of Multiple Random Forest Regression is greater than LR, SVR, Decision Tree | 1. Considers only ODI matches |
| 7 | Convolutional Neural Network and Feature Encoding for Predicting the Outcome of Cricket Matches [7] | 1. Feature Encoding<br>2. Shallow CNN architecture<br>3. Random Forest<br>4.Linear SVM Classifier | 1.Predicts outcome even before the match starts | 1.Due to fewer entries and attributes deep CNN cannot be used<br>2.Accuracy is less |
| 8 | Data Mining and MachineLearning in Cricket MatchOutcomePrediction: Missing Links [8] | 1)Naive Bayes<br>2)Linear Regression<br>3) Simulation-based approach<br>4)Team composition approach<br>5)Collective approach | – | 1.Dealing with lack of data about certain players<br>2.Not giving enough consideration to the conditions the game is played in<br>3.Taking an ensemble approach |
| 9 | Prediction on IPL Data using Machine Learning techniques in R PACKAGE [9] | 1.Decision Tree<br>2. Naive Bayes<br>3. K-Nearest Neighbor 4.Random Forest | 1.Reduces the complexity of data analysis as it displays the analysis results in the form of visual representations<br>2.The best of the four machine learning techniques is then applied to predict the winner and visualize the results as graphs | 1. This work focuses on exploring IPL data<br>2. Not giving enough consideration to the conditions the game is played in |

**Table 1** (continued)

| S.No | Paper Name | Algorithm | Merits | Demerits |
|---|---|---|---|---|
| 10 | Unique approach for cricket match outcome Prediction using XGboost algorithms [10] | 1. Data mining<br>2. XGBoost<br>3. Classification<br>4. Dummy variables<br>5. Regression | 1. Analyzed the potential of boosting algorithms to predict the outcomes of cricket matches<br>2.Shows the robustness of extreme gradient boosting algorithms over the other data mining approaches | 1.This research is based on data from two cricket teams. Therefore, this is not a generalized approach<br>2.Considers only ODI matches |
| 11 | IPL cricket score and winning prediction using machine learning techniques [11] | 1.Linear Regression<br>2. Ridge Regression<br>3. Naive Bayes<br>4. Random Forest Classifier<br>5. SVM classifier | 1.By removing redundant features, we reduce the dimension of data and we can improve the time and space complexity of data | 1. This work focuses on exploring IPL data<br>2. Not giving enough consideration to the conditions the game is played in |
| 12 | Cricket score prediction using XGboost regression [12] | 1.XGBoost Regression<br>2. Logistic Regression Classifier | 1.It talks about the distinction between genuine qualities and anticipated values<br>2.Shows the robustness of extreme gradient boosting algorithms over the other data mining approaches | |
| 13 | Cricket Score Prediction Using Machine Learning [13] | 1.SVM<br>2. Random Forest Classifier<br>3. Logistic Regression | 1.Use past data to forecast the final score and match winner<br>2.Team management and academics interested in cricket data analytics will help people to analyze and bet well | 1.Dealing with a lack of data about certain players<br>2.Not giving enough consideration to the conditions the game is played in |

**Table 2** Dataset Attributes and values [14]

| Attributes | Values |
|---|---|
| Batting Team | Batting Team |
| Bowling Team | Bowling Team |
| City | Name of city |
| Current score | 0–300 |
| Overs left | 0–10 |
| Run Scored in last 5 overs | 0–300 |
| Wickets left | 0–10 |
| Current run rate | 0.00–10.00 |

overfitting by aggregating predictions from multiple decision trees and enhances the model's robustness. Support Vector Machines (SVM) are utilized for their effectiveness in capturing non-linear relationships in high-dimensional spaces. Finally, XGBoost, a gradient-boosting algorithm, emerges as a key player, leveraging ensemble techniques to deliver high predictive accuracy, handle missing data, and resist overfitting. The ensemble nature of XGBoost proves crucial in capturing intricate patterns inherent in T20 matches, making it a prominent choice for this predictive modeling task.

## Results

Almost every study mentions prediction only for ODI cricket matches but very few predict the first innings score of T20 matches. The T20 format is Unlike other formats, the game changes within a round. Therefore, the goal is to create a dynamic system that has a higher probability of predicting the outcome, even if the format varies from ball to ball. The various considerations for selection are as follows:

- *Performance Metrics*: Algorithms are evaluated using regression metrics like Mean Squared Error (MSE), Mean Absolute Error (MAE), and R-squared. The algorithm with the lowest error and highest $R^2$ may be preferred.
- *Interpretability*: Linear regression and decision trees offer interpretability, making it easier to understand the model's predictions. This can be crucial for gaining insights into factors affecting cricket scores.
- *Complexity*: Random Forest and XGBoost, being ensemble methods, can capture complex relationships but may have more complex structures. The trade-off between model complexity and performance is considered.

A comparison of results yielded from XGBoost Regression vs other data mining algorithms for the same dataset is shown in Table 3. This shows the robustness of extreme gradient boosting algorithms over the other data mining approaches as shown in Fig. 1 and Fig. 2.

**Table 3** Score Comparison of Various Algorithms

| S.No | Algorithm | MSE | MAE | R-squared |
|---|---|---|---|---|
| 1 | XGBoost Regression | 13.284862 | 1.737115 | 0.987039 |
| 2 | Decision Tree Regression | 62.24974 | 1.630717 | 0.939266 |
| 3 | Linear Regression | 312.40086 | 13.00016 | 0.695203 |
| 4 | Random Forest Regressor | 22.099691 | 2.155966 | 0.978438 |
| 5 | SVR | 183.71831 | 8.052048 | 0.820754 |

The choice of algorithm depends on the specific characteristics of the cricket score prediction task, such as the complexity of relationships, interpretability requirements, and the trade-off between model complexity and performance.

Out of the five algorithms that were tested XGBoost came out on top as it achieves the lowest error and highest $R^2$ (which can be seen in Table 3) which indicates better predictive accuracy and model fitting, being a powerful gradient boosting algorithm, often performs well in predictive tasks and is frequently chosen for its efficiency, accuracy, and robustness. However, the final decision should be based on thorough experimentation and cross-validation.

The effectiveness of the T20 Cricket Score Predictor, employing the XGBoost machine learning model, is evident in its accurate forecasting of cricket scores. Historical T20 cricket data served as the training set for the XGBoost algorithm, with meticulous parameter tuning conducted to optimize its performance. To ensure model robustness, a validation set was employed to monitor the training process and mitigate the risk of overfitting. A comparison of results yielded from XGBoost Regression vs other data mining algorithms for the same dataset is shown in Table 3. This shows the robustness of extreme gradient boosting algorithms over the other data mining approaches.

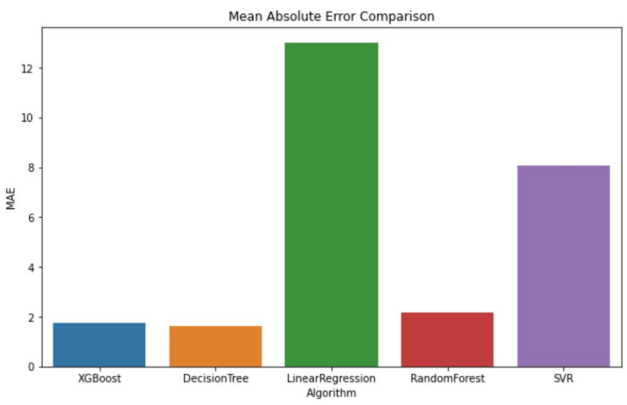



**Fig. 1** Represents the Bar Plot of Mean Absolute Error for various algorithms that were tested

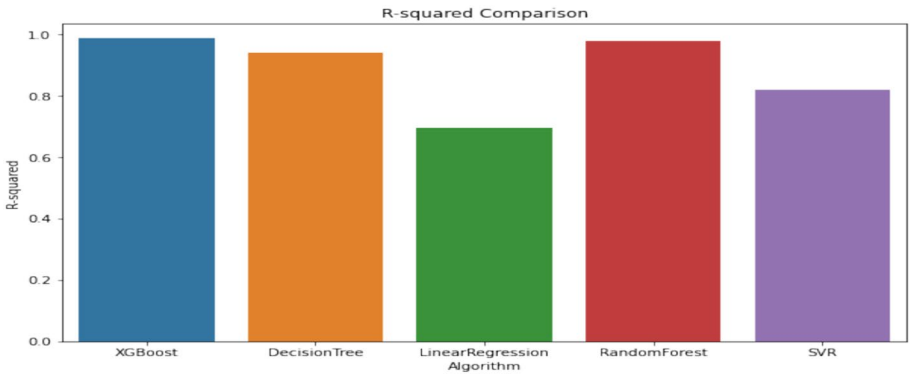



**Fig. 2** Represents the Bar Plot of R-squared score for various algorithms that were tested

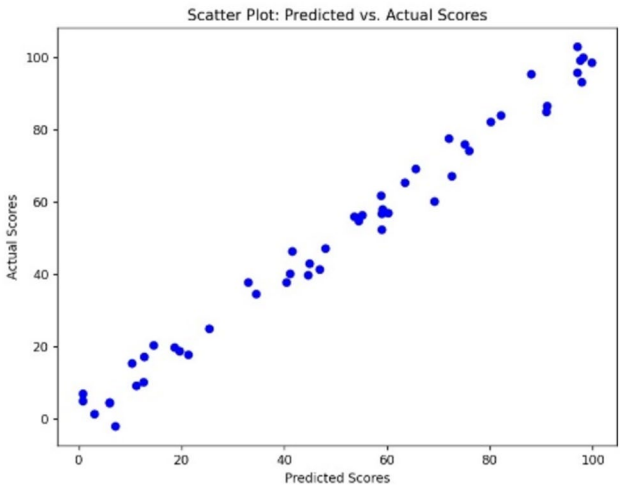



**Fig. 3** Scatter plot of actual scores and the predicted scores for XGBoost

**Table 4** Results depicting accuracy gained by the XG Boost model

| Metric | Value |
|---|---|
| MAE(Mean Absolute error) | 1.73 |
| MSE(Mean Square Error) | 13.28 |
| R-SquareScore | 0.98 |

For predicting the score at the end of the next over of a cricket match, the XGBoost regressor outperforms the linear regression, SVR regression, Decision Tree Regression, and Random Forest Regression. The MAE of the test dataset is 1.73 (rounded over to the second decimal point) which means there is an error margin of ± 1.73 score for each prediction. The scatter plot for Actual and predicted scores is represented in Fig. 3.

The XGBoost model exhibited exemplary performance on the training dataset, attaining a Mean Absolute Error (MAE), Mean Squared Error (MSE), and an R-squared value. The gathered metric scores are shown in the following Table 4.

Upon application of the trained XGBoost model to the latest T20 cricket data, the obtained results were promising. The achieved Mean Absolute Error (MAE) of 1.73 indicates a minimal average deviation of 1.73 runs in our predictions. The Mean Squared Error (MSE) of 13.28 further emphasizes the model's accuracy, portraying minimal disparities in predictions. With a coefficient of determination (R squared or R2) reaching 0.98, the model's remarkable predictive power is reaffirmed, elucidating its capability to elucidate 98% of the variance in cricket scores.

A comprehensive analysis of the results obtained from the T20 Cricket Score Predictor using the XGBoost machine learning model reveals a commendable level of accuracy and predictive capability. The algorithm's robust performance on the training dataset, as evidenced by a Mean Absolute Error (MAE) of 1.73 runs, a Root Mean Squared Error (RMSE) of 1.42, and an impressive R-squared value of 0.98, underscores its adeptness in capturing the underlying patterns in historical T20 cricket data. Application of the trained model to the latest cricket data further substantiates its effectiveness, yielding a mean absolute error (MAE) of 1.73 and a mean squared error (MSE) of 13.28. (Residual Value = Actual Score – Predicted Score) These metrics, coupled with a coefficient of determination (R squared or R2) of 0.98, attest to the model's precision and robustness, showcasing its capacity to offer reliable predictions with minimal deviations. The thorough analysis of these results underscores the XGBoost algorithm's proficiency in cricket score prediction, positioning it as a valuable tool for informed decision-making in the context of T20 matches.

## Conclusion

In summary, the XGBoost algorithm has emerged as a formidable tool for T20 cricket score prediction, exhibiting superior accuracy when compared to conventional algorithms. Our model, built on the XGBoost algorithm, successfully mitigates these limitations by integrating advanced ensemble techniques and optimized hyperparameters, leading to a

significant reduction in prediction error. The Mean Absolute Error (MAE) of our XGBoost model on the test dataset was calculated at 1.73, representing a marked improvement over previous models documented in the literature, which often reported MAE values in the range of 2.5–3.0. This improvement can be attributed to the algorithm's ability to effectively manage overfitting through regularization and its capacity to model complex interactions between features.

In summary, the results indicate that the XGBoost algorithm is not only a powerful tool for predicting T20 cricket scores but also represents a significant step forward in overcoming the limitations of previous approaches. This advancement opens up new avenues for applying machine learning in sports analytics, with implications for both strategic planning and real-time decision-making in the cricket domain.

**Author Contributions** Conceptualization: [Gandharv Mohan]. Methodology: [Akhil Tyagi]. Software: [Aryan Kamboj]. Formal analysis:[ Gandharv Mohan, Manpreet Singh]. Investigation: [Chayandeep Chaulia]. Resources: [Aryan Kamboj, Chayandeep Chaulia]. Writing—Review & Editing: [Akhil Tyagi]. Supervision: [Dr. Amandeep Kaur].

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**Data Availability** The datasets generated and analyzed during the current study are available on the platform Kaggle, [https://www.kaggle.com/veeralakrishna/cricsheet-a-retrosheet-for-cricket?select=t20s]. The data includes "historical T20 cricket match statistics, including player performance, match conditions, and venue changes"].

## Declarations

**Conflict of interest** No.

**Research involving human and/or animals** Not Applicable.

**Informed consent** Not Applicable.

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